



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
S. Y. B.Tech. Civil Engineering

W.E.F. 2026-27

Department of Civil Engineering

Vision

- To excel and lead in education, research, and innovation, contributing to the advancement of design, construction, and maintenance of infrastructure, to enhance the quality of life for humanity in a sustainable way.

Mission

- **M1:** To provide an outstanding learning experience through a rigorous curriculum of theory and practice that develops students' technical and professional skills to succeed in a wide range of careers.
- **M2:** To continually advance research through a culture of discovery, creativity, and innovation to benefit humankind.
- **M3:** To serve as highly capable resources to society, and the profession through professional organizations, consultancy, and continuing education.



Department of Civil Engineering

Program Educational Objectives (PEOs)

- **PEO 1:** Graduate will demonstrate peer-recognized technical competency in the analysis, design, and construction of Civil Engineering Structures.
- **PEO 2:** Graduates will demonstrate leadership and initiative to advance professional and organizational goals with a commitment to ethical standards of profession, teamwork, and respect for diverse cultural backgrounds.
- **PEO 3:** Graduates will be engaged in ongoing learning and professional development through pursuance of higher education and self-study.
- **PEO 4:** Graduates will be committed to create a practice of engineering and other professions in a responsible manner, contributing to the socio-economic development of society.

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics, and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.



WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking, and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior, and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc., with mutual understanding and respect, and of inclusive attitudes.
Program Outcomes (POs)	
PO 1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization as specified in WK1 to WK4 respectively, to develop to the solution of complex engineering problems.
PO 2	Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems, reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO 3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society, and environment as required. (WK5)
PO 4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge, including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO 5	Engineering Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering & IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO 6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO 7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity, and inclusion; adhere to national & international laws. (WK9)
PO 8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.



PO 9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO 10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO 11	Life-long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies, and iii) critical thinking in the broadest context of technological change. (WK8)
Program Specific Outcomes (PSOs)	
• PSO 1: Students will be able to survey, conduct geotechnical investigations, plan, analyze, design, estimate, and construct residences, public buildings, industrial buildings, townships, and infrastructural projects by adopting appropriate construction methods. • PSO 2: Students will be able to analyze and design the water resources systems, and municipal and industrial waste treatment plants, with due consideration to a pollution-free environment. • PSO 3: Students will be able to use appropriate application software, develop skills necessary for professional practice as a Civil Engineer, and prepare themselves for competitive examinations for higher education & for public service commissions.	



Department of Civil Engineering

Legends Used

L	Lecture Hours/week
T	Tutorial Hours/week
P	Practical Hours/week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology



Department of Civil Engineering

Course Code Format

2	5	C	E	U/P	2	C	C	1	T/L
Year of Syllabus revision		Program Code		U-Under Graduate P-Post Graduate	Semester No./ Year1/2/3/...8	CourseType		Course Serial No. 1-9	T-Theory, L-Lab session A-Tutorial P-Programming / Design

Program Code

CE Civil Engineering

Course Type

BS Basic Science

ES Engineering Science

HU Humanities & Social Science

MC Mandatory Course

CC Program Core Compulsory Course

SN* Self-Learning (N* indicates the serial number of electives offered in the respective category)

EN* Program Core Elective Course (N* indicates the serial number of electives offered in the respective category)

SK Skill-Based Course

SM Seminar

MP Mini project

PR Project

IN Internship

ON* Open Elective (N* indicates the serial number of electives offered in the respective category)

MD Multidisciplinary Minor

EM Entrepreneurship/Economics/Management Courses

FP Community Engagement Project / Field Project



AE	Ability Enhancement Course
VE	Value Education Course
IK	Indian Knowledge System
VS	Vocational & Skill Enhancement Course
RM	Research Methodology
HN	Honors' Degree Course
HR	Honors Research

Sample Course Code	
25CEU5CC1T	Design of Steel Structures



Department of Civil Engineering

B. Tech. Semester III

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
25CEU3CC1T	Fluid Mechanics and Fluid Machines	2	-	-	2	60		40	-	100
25CEU3CC2T	Building Construction & Design	3	-	-	3	60		40	-	100
25CEU3CC3T	Structural Mechanics-I	2	-	-	2	60		40	-	100
25##U3ON*4T	Open Elective I	2		-	2	60		40	-	100
25##U3ON*4A	Open Elective I (Tutorial)	-	1		1	-		-	25	25
25CMU3EM5T	Entrepreneurship development	1	1	-	2	-		-	50	50
25CEU3MD6T	Multidisciplinary Minor-I	2	-	-	2	60		40	-	100
25CMU3VE7T	Universal Human Values	2	-	-	2	50*		-	-	50
25CEU3CC8T	Surveying & Geomatics	2			2	60		40		100
	Subtotal	16	2	-	18	410		240	75	725
Laboratory Courses										
25CEU3CC1L	Fluid Mechanics and Fluid Machines Lab	-	-	2	1		25		25	50
25CEU3CC8L	Surveying & Geomatics Lab	-	-	2	1		25		25	50
25CEU3FP9L	Community Engagement Project / Field Project	-	-	4	2		-		50	50
	Subtotal	-	-	8	4	-	50		100	150
	Grand Total	16	2	8	22	410	50	240	175	875

Note:

- N* indicates the serial number of electives offered in the respective category
- ## indicates program code of the offering program
- *Examination will be MCQ-based.



Department of Civil Engineering

B. Tech. Semester III

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/POE	ISE	ICA	
25CEU3CC1T	Fluid Mechanics and Fluid Machines	2	-	-	2	60		40	-	100
25CEU3CC2T	Building Construction & Design	3	-	-	3	60		40	-	100
25CEU3CC3T	Structural Mechanics-I	2	-	-	2	60		40	-	100
25##U3ON*4T	Open Elective I	2		-	2	60		40	-	100
25##U3ON*4A	Open Elective I (Tutorial)	-	1		1	-		-	25	25
25CMU3EM5T	Entrepreneurship development	1	1	-	2	-		-	50	50
25CEU3MD6T	Multidisciplinary Minor-I	2	-	-	2	60		40	-	100
25CMU3VE7T	Universal Human Values	2	-	-	2	50*		-	-	50
25CEU3CC8T	Surveying & Geomatics	2			2	60		40		100
	Subtotal	16	2	-	18	410		240	75	725
Laboratory Courses										
25CEU3CC1L	Fluid Mechanics and Fluid Machines Lab	-	-	2	1		25		25	50
25CEU3CC8L	Surveying & Geomatics Lab	-	-	2	1		25		25	50
25CEU3FP9L	Community Engagement Project / Field Project	-	-	4	2		-		50	50
	Subtotal	-	-	8	4	-	50		100	150
	Grand Total	16	2	8	22	410	50	240	175	875

Note:

- N* indicates the serial number of electives offered in the respective category
- ## indicates program code of the offering program
- *Examination will be MCQ-based.

Department of Civil Engineering										
B. Tech. Semester IV										
Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
25CEU4CC1T	Environmental Engineering	2	-	-	2	60	-	40	-	100
25CEU4CC2T	Concrete Technology & Material Testing	2	-	-	2	60	-	40	-	100
25CEU4CC3T	Structural Mechanics-II	2	-	-	2	60	-	40	-	100
25CEU4CC3A	Structural Mechanics-II-Tutorial	-	1	-	1	-	-	-	25	25
25CEU4EM4T	Principles of Management and Quantitative Techniques	2	-	-	2	60	-	40	-	100
25##U4ON*5T	Open Elective II	2	-	-	2	60	-	40	-	100
25##U4ON*5A	Open Elective II (Tutorial)	-	1	-	1	-	-	-	25	25
25CEU4MD6T	Multidisciplinary Minor II	1	-	-	1	-	-	50	-	50
25CMU4AE7T	General Proficiency	1	1	-	2	-	-	-	50	50
25CMU4VE8T	Professional Ethics	2	-	-	2	-	-	50	-	50
Subtotal		14	3	-	17	300	-	300	100	700
Laboratory Courses										
25CEU4CC1L	Environmental Engineering Lab	-	-	2	1	-	25	-	25	50
25CEU4CC2L	Concrete Technology & Material Testing Lab	-	-	2	1	-	-	-	25	25
25CEU4MD6L	Multidisciplinary Minor II Lab	-	-	2	1	-	-	-	25	25
25CEU4CC9L	Building Planning and Modelling Lab	1	-	2	2	-	25	-	25	50
25CEU4PC10P	Programming with Python	1	-	2	2	-	25	-	50	75
Subtotal		2	-	10	7	-	75	-	150	225
Grand Total		16	3	10	24	300	75	300	250	925

- **Mandatory Course:** The Environmental Studies course will be taught in both Semester III and IV, whereas the assessment will be in Semester IV as an End Semester Examination.

CourseCode	NameofCourse	Engagement Hours			Credits	SA	FA		
		L	T	P			ESE	ISE	ICA
25CMU4MC2T	Environmental Studies	1	-	-	-	50	-	-	50

Note:

- *Examination will be MCQ-based.
- N* indicates the serial number of electives offered in the respective category
- ## indicates the program code of the offering Program
- Internal Continuous Assessment (ICA): ICA shall be a continuous process based on the performance of the student in assignments, class tests, quizzes, attendance and interaction during theory and lab sessions, journal writing, report presentation, etc., as applicable.
- Students must study and pass Environmental Studies in the Second Year of B. Tech. Civil Engineering to become eligible for the award of a degree.

List of Open Electives:

• **Open Electives I-(Semester-III)**

Sr.No.	Subject Code	Subject
1	25CEU3O14T	Managerial Economics
2	25MAU3O24T	Renewable Energy
3	25ETU3O34T	Sustainable Development
4	25CSU3O44T	Management Information Systems
5	25ECU3O54T	Fundamentals of Digital Marketing
6	25ITU3O64T	Cyber Laws

Open Electives II (Semester-IV)

Sr.No.	Subject Code	Subject
1	25GEU4O15T	Higher Engineering Mathematics
2	25GEU4O25T	Advanced Engineering Mathematics
3	25GEU4O35T	Applied Mathematics
4	25GEU4O45T	Statistics and Fuzzy Logic
5	25GEU4O55T	Applied Statistics
6	25GEU4O65T	Computational and Optimization Techniques

● **For 25MCU5OE7T: Open Elective III (MOOC) in Semester V:**

1. Students should to enroll in one of the courses of a minimum duration of 8 weeks offered on the SWAYAM/NPTEL platform. The list of courses will be finalized and released by the Board of Studies each year.
2. A list of MOOCs will be provided by the department, depending on the availability of the courses in that semester under NPTEL / Swayam or Other recognized MOOC Platforms as per suggestions by the BoS.
3. Students may enroll for the course in Semester III, IV or V. They must complete all assignments and appear for the certification examination conducted by SWAYAM/NPTEL.
4. Students must pass the examination by the end of Semester V. The marks earned by the student in the final assessment of this MOOC will be appropriately scaled and transferred to **Open Elective III (MOOC)** in Semester V.

● **Multidisciplinary Minor (MDM) Courses**

Students can choose one of the following Multidisciplinary Minor Program during the start of Semester III.

Sr. No	MDM Program	MDM I (Sem III)	MDM II (Sem IV)	MDM III (Sem V)	MDM IV (Sem VI)	MDM V (Sem VII)
1	Computer Science and Engineering	Operating System	UI Technologies	Software Engineering	Big Data Technologies	Software Testing and Quality Assurance
2	Information Technology	Principles of Operating Systems	Web UI and UX Technology	Software Engineering Principles	Data Preprocessing and Visualization	Cyber Security
3	Mechanical and Automation Engineering	Manufacturing Processes and Mechanisms	Machine Drawing and 3D Modeling	Automotive Engineering and Robotics	Additive Manufacturing	Thermal Systems
4	Electronics and Telecommunication Engineering	Fundamentals of Electronic Circuits	Electronics Design and Prototyping	Introduction to Embedded Systems	Fundamentals of Communication Techniques	Enclosure and Communication Design for IoT
5	Electronics and Computer Engineering	Analog Electronics	Digital Electronics	Microcontrollers and Peripherals	Advanced Controllers and Interfacing	Electronic System Design
6	Artificial Intelligence and Machine Learning	Fundamentals of Machine Learning	Data Science and Visualization	Reinforcement Learning	Applied Machine Learning	Generative AI



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Civil Engineering), Semester-III

25CEU3CC1T: Fluid Mechanics and Fluid Machines

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
		POE	25 Marks
Introduction:			
The course offers a comprehensive introduction to the principles of fluid mechanics and the design and operation of fluid machines. Students will explore fluid properties, fluid statics, and fluid dynamics, learning to analyze fluid flow in various systems. The course covers essential topics such as laminar and turbulent flow, flow through pipes, and the principles and performance of fluid machinery, including pumps, turbines, and compressors.			
Course Prerequisite:			
Basics of Engineering Physics, Engineering Mechanics, and Mathematics.			
Course Objectives:			
1. To make students develop a fundamental understanding of the physical properties of fluids and their behavior under various conditions. 2. To equip students with the ability to analyze and solve problems related to fluid dynamics, including laminar and turbulent flow, boundary layers, and flow through pipes and channels. 3. To enable students to design, evaluate, and select appropriate fluid machinery such as pumps, turbines, and compressors. 4. To provide students with hands-on experience and practical skills through laboratory experiments and real-world case studies.			
Course Outcomes:			
After completing the course, students will be able to 1. Comprehend and analyze the physical properties of fluids, hydrostatic forces on submerged and floating bodies, and their interrelationships. 2. Analyze fluid motion under static and dynamic conditions, including flow behavior with and without external forces. 3. Evaluate and quantify water flow through notches, weirs, and pipe systems, including determination of head losses and analysis of pipe networks. 4. Describe the working principles and assess the performance parameters of hydraulic turbines and pumps.			



Unit – I	Fluid Properties	4 Hours
Definition of Fluid and fluid mechanics, Physical properties of fluids: Density, Specific weight, Specific volume, Relative density, Cohesion, Adhesion, Viscosity- Newtonian and Non-Newtonian fluids, Classification of fluids, compressibility, Surface tension, Capillarity, Vapor pressure, Cavitation, Problems involving use of above Fluid Properties.		
Unit – II	Fluid Statics	6 Hours
Pascal's law, Hydrostatic Law, units and scale of pressure measurement, types of pressure, Piezometer, U-tube manometer, U-tube differential manometer, Inverted U-tube differential manometer; Mechanical pressure gauge – Bourdon tube, Total Pressure on vertical and inclined plane surfaces submerged in fluid, depth of Center of Pressure. Concepts of: Buoyant force, Buoyancy, Center of Buoyancy, Archimedes Principle, Metacentre, and Metacentric height, Equilibrium of floating and submerged bodies		
Unit – III	Fluid Kinetics & Fluid Dynamics:	6 Hours
Methods of describing fluid motion, Concept of Control Volume, Streamlines, Path lines, Streak lines and Stream tubes, Types of fluid flows, Rate of flow, Continuity equation, Velocity potential and stream function, Equipotential Line, flow net. Forces acting on fluid mass in motion, Euler's equation of motion along a streamline, Bernoulli's Theorem, Limitations and Applications, Measurement of discharge through Venturi meter, Orifice meter, and Orifice, Measurement of velocity through Pitot tube.		
Unit – IV	Notches and Weirs	4 Hours
Notches: Types of notches, Derivation of discharge equation, velocity due to approaches, Francis formula, Errors in measurements of head. Weir: Sharp & Broad crested weirs, calibration of weirs, time required to empty the tank with notches and weirs, Ventilation of weirs, Types of nappe.		
Unit – V	Flow Through Pipes	4 Hours
Energy Losses in pipe flow: Major and Minor Losses, Darcy Weisbach Equation, Pipes in Series and Parallel, Concept of Equivalent length and Equivalent diameter of pipe; Concept of HGL, TEL and Syphon, Water Hammer; Hardy Cross Method for solving pipe network.		
Unit – VI	Impact of Jet, Hydraulic Turbines, and Pumps	6 Hours
Force and work done due to impact of jet on stationary and moving, flat and curved surfaces using linear momentum principle, Elements of hydropower plant, Hydraulic turbines- Classification, Francis and Kaplan turbines, Draft tube, Specific speed, Design and governing of Pelton Wheel, Prediction of performance in terms of unit quantities and specific quantities, selection of turbines based on head and specific speed, Classification of Hydraulic Pumps, Centrifugal Pump - Components, Working of Centrifugal pump, inlet and outlet velocity triangles, Work done by impeller, Types of Heads and		



efficiencies, Concept of priming of pump, multistage pumping, operational difficulties in Centrifugal Pump, Introduction to submersible pumps and reciprocating pumps, Selection of Pumps.

INTERNAL CONTINUOUS ASSESSMENT (ICA)

(A) Experiments (Min.10):

1. Measurement of pressure (Piezometer, Manometers, Pressure gauges)
2. Determination of Metacentric height
3. Verification of Bernoulli's Theorem
4. Calibration of an orifice/mouthpiece.
5. Calibration of the Venturi meter.
6. Calibration of V-notch and rectangular notch
7. Calibration of the broad and sharp-crested suppressed weir
8. Calibration of Ogee Weir.
9. Determination of loss of head in pipe flow.
10. Reynolds Experiment
11. Impact of jet
12. Study of turbines (demonstration/test)
13. Study of centrifugal pump. (demonstration/test)

Internal Continuous Assessment (ICA) submission shall consist of a journal containing

1. Above mentioned Experiments
2. Assignments on each unit based on the syllabus.

Text Books

1. Fluid Mechanics -A. K. Jain-Khanna Pub., Delhi.
2. Fluid Mechanics - S. Nagrathanam Khanna Pub., Delhi.
3. Fluid Mechanics - Garde, Mirajgaonkar Nemchand and Bros., Roorkee.
4. Fluid Mechanics - R. W. Fox, P.J. Prichard, A. T. McDonold- Wiley India.
5. Fluid Mechanics - K.L. Kumar Eurasia Publishing House, Delhi.
6. Fluid Mechanics & Hydraulic Machines, SS Rattan, Khanna Publishing House

Reference Books

1. Fluid Mechanics and Fluid Machines - Modi and Seth, Standard Book House, Delhi.
2. Fluid Mechanics - Streeter McGraw Hill-International Book Co., Auckland.
3. Fluid Mechanics - Munson, Young- Willy India.
4. Mechanics of Fluids - M.C. Potter, Wiggert, Ramdan- Cengage Learning
5. Elementary Fluid Mechanics - H. Rouse Toppan C. Ltd., Tokyo.
6. Fluid Mechanics – Shames McGraw-Hill International Book Co.

e-Resources

1. NPTEL (National Programme on Technology Enhanced Learning) – Fluid Mechanics
URL: <https://nptel.ac.in/courses/112/105/112105171/>
2. NPTEL – Fluid Machines-URL: <https://nptel.ac.in/courses/112/105/112105267/>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Civil Engineering), Semester-III

25CEU3CC2T: Building Construction and Design

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Credits	3	ISE	40 Marks
Introduction:			
This course provides a thorough introduction to the functional requirements and structures of buildings. It also covers the development of CADD skills for drawing building components, creating detailed perspective view drawings, and planning and designing residential buildings with CADD to ensure compliance with building bylaws.			
Course Prerequisite:			
Basic knowledge of Building and its various components.			
Course Objectives:			
1. To make students understand the functional requirements of building structures. 2. To make students develop CADD skills for building components. 3. To make students create detailed perspective view drawings 4. To make students plan and design residential buildings with CADDwith building bylaws. 5. To ensure that students about the building services, rainwater harvesting, and electrification concepts.			
Course Outcomes:			
After completing the course, students will be able to			
1. Elucidate the functional requirements of buildings, types of foundations, and types of Masonry walls. 2. Draw neat drawings of different building components, such as doors, windows, stairs, etc., with a suitable scale using CADD software. 3. Draw neat perspective view drawings of an object. 4. Apply the site selection criteria for building, Principles of Building Planning, and Byelaws for residential building planning and design. 5. Apply building services, rainwater harvesting, and electrification concepts as per requirements for residential buildings.			
Unit – I	Building functional Requirements, Building Type, Foundation & Type of Masonry wall		8 Hours
Building functional Requirements - Strength, Stability, Comfort, Convenience, Safety, Damp Prevention, Water Proofing, Heat Insulation, Day Lighting, Ventilation, Termite Resistance.			
Building Types – Framed, Load Bearing and Composite structures, Comparison between all three types.			



Building components (elements) Foundation: - Importance of foundation as a transferring building element. Shallow Foundations – Wall footing, Isolated footing, Combined Footing, Strap Footing, Continuous or Strip Footing, Cantilever Footing, Raft Foundation. (Reinforcement placement not expected) Types of Masonry and Walls: Introduction to Scale and various types of Scale. Introduction to Stone masonry walls, bonding, and breaking of Joints. Brick masonry walls – Standard Brick size and Properties of good brickwork. Bonds-Stretcher, Header, Flemish & English bond (up to 1 ½ Brick thick)		
Unit – II	Doors, Windows & Staircase	8 Hours
Door Types: - Paneled, Flush, Glazed. Door elements, Fixtures, and Fastenings. Window: - Types: Steel Glazed, Wooden Paneled, Aluminum Glazed Sliding Ventilators, and Fixed Glass windows. Staircase: - Functional requirements of the stairs, design of stairs, types of stairs, technical terms		
Unit – III	Perspective Drawing	6 Hours
Elements of Perspective drawings, parallel perspective (One Point), and angular perspective (Two Point) drawing. Perspective drawing of objects and perspective drawing of one G+1 Residential building (Readymade plan to be given to the students).		
Unit – IV	Site Selection of Building, Principles of Building Planning, Orientation and By-Laws, and Dimension Relationships	6 Hours
Site selection criteria for building. Principles of Building Planning and the significance of Sun Diagram (Sun Path Diagram) and Wind Flow Direction. Orientation: - Basic Zones of India based on climate conditions, Orientations of buildings for various parts of India based on climate conditions. Building Planning Byelaws and Regulations as per SP-7, National Building Code of India. Dimensions & Space requirement with body measurements. Space design for passage between walls, service access, stairs, ramps, and elevators.		
Unit – V	Planning and Design of Residential Buildings	6 Hours
Planning and functional requirements of Residential buildings: - Bungalows (Detached), Twin bungalows (Semi-Detached), Rowhouses, Ownership flats, and Apartments, Parking Area Criteria		
Unit – VI	Building Services	6 Hours
Plumbing Systems: - Significance of Plumbing and Drainage plan and layout, Water Supply Requirements for Buildings, various types of traps, Fittings, Chambers, and various types of materials like PVC, GI, AC, CI, HDPE, Stoneware, CPVC with various gauges and thickness, Water Closet Pan: Types and size. Introduction to the Concept and Design of Rainwater Harvesting. Electrification: - Concealed and open wiring systems, requirements, and locations of various Electrical points, Concept of earthing		



INTERNAL CONTINUOUS ASSESSMENT (ICA)

A. Sketching in a sketchbook consisting of the following drawing exercises: (To Scale)

1. Lettering, Symbols, and line work.
2. Framed and Load-Bearing Building Structure.
3. Types of Foundations.
4. Bricks: Bats and Closers
5. Brick bonds: Stretcher, Header, Flemish, and English bond.
6. Doors.
7. Windows.
8. Staircases.
9. Perspective drawing of the object.

B. Sketching in a sketchbook consisting of the following drawing exercises: (Not to Scale)

1. Various types of traps.
2. Single-Stack System.
3. One-Pipe System
4. One-Pipe System with partial ventilation.
5. Two-Pipe System.

C. Drawing using CADD software (To Scale):

1. Doors.
2. Windows.
3. Staircases.

Remark: Prints of these CADD drawings should be to the Scale, which will be a part of “Term work”.

Text Books

1. A textbook of Building Construction- Arora & Bindra- Dhanpat Rai Publication, New Delhi.
2. Building Construction- Sushil Kumar- Standard Publishers, Delhi.
3. Building Construction – Arora & Gupta –Satya Prakashan, New Delhi.
4. Principles of Building Drawing- M.G. Shah and C.M. Kale.
5. A course in Civil Engineering Drawing- V.B. Sikka – S.K.Katariya & Sons, Delhi.
6. Civil Engineering Construction Materials, S.K. Sharma, KBP House
7. Engineering Drawing + AutoCAD, by K.Venugopal, New Age International Publishers
8. Mastering AutoCAD 2019 and AutoCAD LT 2019, George Omura and Brian C. Benton, SYBEX Publishers.

Reference Books

1. Building Technology- Ivor H. Seely.



2. Building Construction-Makay vol. I & II
3. National Building Code of India-SP7- Indian Standards, Delhi.
4. Various IS Specifications for Drawings, Symbols, Conventional Signs as per IS 962-1967-Indian Standards, Delhi.
5. Building Construction, A to Z – Mantri.
6. Building Materials- TTTI, Chandigarh.
7. Building Construction- S.S. Bhavikatti- Vikas Publishing House Pvt. Ltd., Noida.
8. Building Materials- S.S. Bhavikatti- Vikas Publishing House Pvt. Ltd., Noida
9. A Course in Civil Engineering Drawing, Sikka, S.K., Kataria & Sons
10. Engineering Drawing, Dhanarajay A. Jolhe, Tata McGraw-Hill
11. Engineering Drawing + AutoCAD, by K.Venugopal, New Age International Publishers
12. Mastering AutoCAD 2019 and AutoCAD LT 2019, George Omura and Brian C. Benton, SYBEX Publishers.

e-Resources

1. NPTEL – Building Materials and Construction-URL:
<https://nptel.ac.in/courses/105/102/105102088/>
2. Wikibooks – Building Construction- https://en.wikibooks.org/wiki/Building_Construction





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Second Year B.Tech. (Civil Engineering), Semester-III

25CEU3CC3T: Structural Mechanics-I

Teaching Scheme

Lectures 2 Hours/week

Credits 2

Teaching Scheme

ESE 60 Marks

ISE 40 Marks

Introduction:

This course covers the fundamentals of structural materials and material behaviour essential for engineering design. Topics include stress-strain relations, principal stresses, shear force diagrams, bending moment diagrams, and influence line diagrams, emphasising practical applications in engineering practice

Course Prerequisite:

The student has completed a course in Applied Mechanics. Prior knowledge of statics, mechanics of materials, and basic calculus is essential. Familiarity with concepts such as equilibrium of forces and material properties will facilitate understanding of the topics.

Course Objectives:

1. To explain the behaviour of engineering materials under axial loading by interpreting simple stresses and strains, Hooke's law, elastic constants, and fundamental stress-strain concepts
2. To analyse principal stresses and strains in 2-D stress systems and to evaluate major theories of elastic failure for predicting material behaviour under complex loading.
3. To apply the fundamental relationship between loading, shear force and bending moment in order to construct and interpret shear force and bending moment diagrams for beams.
4. To apply the principles of simple bending theory and flexure formula to determine the bending stresses and shear stresses in beams of various cross sections.
5. To analyse statically determinate beams using influence lines and the Muller Breslau principle for evaluating reactions, shear force, and bending moments under moving loads

Course Outcomes:

After completing the course, students will be able to

1. Describe and interpret stress-strain behaviour to illustrate the basic responses of material in practical engineering scenarios.
2. Identify principal planes & determine principal stresses and strain in 2-D loading conditions and assess the applicability of various elastic failure theories to predict and prevent material failure in engineering applications



3. Calculate shear forces and bending moments to construct the corresponding diagrams for beams under various loading conditions. 4. Evaluate bending stresses, moment of resistance, and shear stress distribution in beams with various cross-sectional geometries for effective structural assessment. 5. Analyse, interpret and construct influence lines for beams and apply knowledge of influence line diagrams for determining critical values related to structural design		
Unit – I	Simple stresses and strains	6 Hours
The behaviour of Engineering materials under axial loading, Simple stresses and strains, Hooke's law, Stress-strain relations for ductile and brittle material, elastic constants, working stress, Factor of safety, Stresses & strains in three dimensions (linear, lateral, shear, and volumetric), normal and shear stresses, Complementary shear stress, the relation between elastic constants, the assumption in elastic analysis, Composite sections under axial loading: compound bars.		
Unit – II	Principal stresses and strains	5 Hours
Principal stresses and strains for 2-D Problems-Normal and shear stress on an inclined plane. Principal plane and Principal stresses, Principal strains, Principal stresses in beams		
Unit – III	Theories of elastic failures	4 Hours
Maximum principal stress, Maximum principal strain, Maximum shear stress, Total strain energy, and distortion energy.		
Unit – IV	Analysis of statically determinate beams	5 Hours
Shear force and bending moment diagrams for beams subjected to point load (inclined load also), uniformly distributed load, uniformly varying load, and couples. Relation between the intensity of load, shear force, and bending moment		
Unit – V	Bending Stresses	3 Hours
Simple bending theory, pure bending of beams, flexure formula, the moment of resistance of different cross sections, built-up sections, Rectangular, Circular, and flanged sections.		
Unit – VI	Shear stresses in beams	3 Hours
Distribution of shear stresses in beams of various commonly used sections such as rectangular, triangular, circular, T, and I sections.		
Unit – VII	Influence line diagrams	4 Hours
Influence line diagrams, Muller-Breslau principle, Application to statically determinate simple and compound beams to determine support reaction, S.F. & B.M. at any section.		



Text Books
1. Strength of Materials by F.L. Singer, Harper and Row Pub., New York. 2. Strength of Materials by R.K.Bansal, Laxmi Publications 3. Strength of Materials by Bhavikatti, Vikas Publications, New Delhi. 4. Strength of Materials by Ramamurtham, Dhanpat Rai & Sons, New Delhi 5. Strength of Materials by R. S. Khurmi, S. Chand Publication, New Delhi 6. Mechanics of Structures (Vol. I & II) by S. B. Junnarkar, Charator Book House, Anand. 7. Strength of Materials by R. K. Rajput, S. Chand Publication, New Delhi 8. Strength of Materials, D.S. Bedi, Khanna Publishing House 9. Strength of Materials, R.K. Bansal, Laxmi Publications
Reference Books
1. Analysis of Structures (Vol- I& II) by Vazirani and Ratwani, Khanna Pub., Delhi. 2. Structural Analysis by C. S. Reddy, Tata Mc. Graw Hill, New Delhi. 3. Introduction to Mechanics of Solids by E. P. Popov. Prentice- Hall of India. 4. Elementary Structural analysis by Norris and Wilbur Mc-Graw Hill, New York. 5. Elements of Strength of Materials, (Recent Edition) by S. Timoshenko and J. Young Affiliated East-West 6. Mechanics of Materials by Gere and Timoshenko, C.B.S. Delhi 7. Strength of Materials, R. Subramanian, Oxford University Press
e-Resources
1. https://www.youtube.com/playlist?list=PLOQ_D0oq27oCPBudQXnRf8G4UNQAOpEcv 2. https://nptel.ac.in/courses/105105108



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Civil Engineering), Semester-III****25CMU3EM5T: Entrepreneurship Development**

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ESE	-
Tutorial	1 Hour/week	ISE	-
Credits	2	ICA	50 Marks
Introduction:			
Entrepreneurship education in India has gained relevance in today's context. Education in the area of entrepreneurship helps students develop skills and knowledge, which could benefit them in starting, organizing, and managing their enterprises. Entrepreneurship education encourages innovation, fosters job creation, and improves global competitiveness. This course will focus on key attributes of entrepreneurship, including the qualities of a successful entrepreneur, Entrepreneurship Development Programmes, Ideation Techniques, Business Plan Formulation, and Various Support Systems. To sum up, the course will enable students to have an understanding of the complete entrepreneurial ecosystem.			
Course Prerequisite:			
This course requires familiarity with basic business terminology, management principles, and economics.			
Course Objectives:			
1. To familiarize the students with entrepreneurship and its significance in national development 2. To acquaint the students with the options available to new entrepreneurs 3. To formulate a business plan/project report for a startup 4. To acquaint the students with the support system associated with entrepreneurial development			
Course Outcomes:			
At the end of the course, the students will be able to: 1. Identify characteristics and types of entrepreneurs. 2. Analyze the significance of Entrepreneurship Development Programmes (EDPs), their structure, phases, and the challenges faced in their implementation. 3. Demonstrate the ability to generate business ideas and prepare a comprehensive entrepreneurial project report. 4. Compare different structures, ownership forms, funding options and interpret the role of government support systems in promoting entrepreneurship.			
Unit – I	Entrepreneur	3 Hours	



Concept, meaning, and definitions of entrepreneur, need of entrepreneur, intrapreneur, social entrepreneur, qualities of entrepreneurs, and types of entrepreneurs.		
Unit – II	Entrepreneurship Development	4 Hours
Concept of entrepreneurship, Entrepreneurship Development Programmes (EDPs)- meaning & need of EDPs, course content & curriculum of EDPs, phases of EDPs, problems of EDPs		
Unit – III	Entrepreneurial Project Development	4 Hours
Idea generation–sources and methods, preparation of a project report/ business plan including: market plan, financial plan, operational plan, HR plan, working capital management, break-even analysis etc.		
Unit – IV	Small-Medium Enterprises and Support Systems	4 Hours
Meaning and definition of Micro, Small & Medium Enterprises, forms of business ownership, Funding options available, role of government organizations to support business.		
INTERNAL CONTINUOUS ASSESSMENT (ICA) Students of a batch should be divided into groups (consisting of a maximum of five members) to carry out the following tasks: 1. Two case studies on successful entrepreneurs 2. Two case studies on the failure of businesses 3. Idea generation & selection of an idea for a business 4. Preparation of project report/business plan for starting a small unit and presentation on the same.		
Text Books		
1. Entrepreneurial Development, Dr. S. S. Khanka, S. Chand Publications 2. Dynamics of Entrepreneurial Development and Management - Vasant Desai, Himalaya Publishing House		
Reference Books		
1. Entrepreneurship & Small Business, Michael Schaper, Thierry Volery, Pauli Weber, Kate Lewis, Wiley Publication 2. Entrepreneurship, Robert Hisrich, Michael Peters, Dean Shepherd, Sabyasachi Sinha, McGraw-Hill Publication 3. Entrepreneurship, Alpana Trehan, Dreamtech Press 4. Small-Scale Industries and Entrepreneurship - Vasant Desai, Himalaya Publishing House		
E-resources:		
https://archive.nptel.ac.in/courses/127/105/127105007/		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Civil Engineering), Semester-III****25CMU3VE7T: Universal Human Values**

Teaching Scheme		Examination Scheme	
Tutorials	2 Hours/week	ESE	50 Marks
Credits	2		
Introduction:			
This course transitions from "information-based" education to "value-based" education. It facilitates a process of Self-Exploration to help students develop a holistic perspective of life, profession, and society. For Engineering students, this serves as a foundation for ethical infrastructure development and sustainable urbanization.			
Course Prerequisite:			
Basic knowledge of Engineering Mathematics, Physics, and Basic Civil Engineering.			
Course Objectives:			
This introductory course input is intended: 1. To facilitate a holistic perspective toward life and profession based on correct understanding of Human Reality. 2. To understand the complementarity between Values and Skills (Right Understanding vs. Technical Competence). 3. To enable students to recognize the harmony in Self, Family, Society, and Nature. 4. To internalize ethical conduct and professional integrity as per NBA PO8 (Ethics) and PO7 (Sustainability).			
Course Outcomes:			
At the end of the course, students will be able to 1. Apply the process of self-exploration to distinguish between "Human Aspirations" and "Physical Facilities". 2. Analyze the coexistence of the 'I' and the 'Body' to maintain health and mental harmony. 3. Evaluate the values required for trustful and mutually fulfilling human relationships in family and society. 4. Design sustainable solutions by understanding the interconnectedness and recyclability in Nature.			
Unit – I	Introduction to Value Education & Self-Exploration		7 Hours
NEP 2020 Perspective: Need for value-based education in technical institutes. Process of Self-Exploration: Natural Acceptance vs. Peer Pressure; Experiential Validation.			



Basic Human Aspirations: Continuous Happiness and Prosperity. The Trinity: Right Understanding, Relationship, and Physical Facilities (Correct Priority). Current Scenario: Critical appraisal of over-consumption vs. sustainability.		
Unit – II	Harmony in the Human Being	7 Hours
The Co-existence: Understanding the Self ('I') as sentient and the Body as material. Needs Analysis: Sukh (Happiness) vs. Suvidha (Physical Facilities). The Self as the Doer, Seer, and Enjoyer: Understanding the activities of 'I'. Harmony of 'I' with Body: Sanyam (Self-regulation) and Swasthya (Health). Professional Application: Impact of stress and lifestyle on engineering productivity.		
Unit – III	Harmony in Family and Society	8 Hours
Values in Relationships: Trust (Vishwas) and Respect (Samman) as foundational values. Justice (Nyaya): Ensuring mutual fulfillment (Ubhay-tripti) from family to world family (Vasudhaiva Kutumbakam). Social Harmony: Samadhan, Samridhi, Abhay, Sah-astitva (Resolution, Prosperity, Fearlessness, Co-existence). Societal Extension: Transition from Akhand Samaj (Undivided Society) to Sarvabhaum Vyawastha (Universal Order).		
Unit – IV	Harmony in Nature & Universal Order	8 Hours
The Four Orders of Nature: Material, Bio, Animal, and Human Order. Interconnectedness: Recyclability and self-regulation in the environment. Existence as Co-existence: Submergence of units in Space. Sustainability for Civil Engineers: Understanding "Construction" within the limits of "Nature's Order." Holistic Perception: Developing a "Green Perspective" in infrastructure.		
INTERNAL CONTINUOUS ASSESSMENT The Universal Human Values (UHV) course emphasizes introspection, values, and ethical thinking, many of which are affective domain outcomes and not as easily measurable as cognitive skills.		
Text Books		
1. R.R. Gaur, R. Sangal, G.P. Bagaria, A Foundation Course in Human Values and Professional Ethics, Excel Books, 2010. 2. Teachers' Manual: A Foundation Course in Human Values and Professional Ethics, Excel Books, 2010.		
Reference Books		
1. A. Nagraj, Jeevan Vidya: Ek Parichay, Divya Path Sansthan. 2. E.F. Schumacher, Small is Beautiful: A Study of Economics as if People Mattered. 3. A.N. Tripathy, Human Values, New Age International Publishers. 4. S.D. Sharma, Indian Ethos and Modern Management.		



e-Resources:
1. UHV Official Website 2. AICTE Online Workshops on UHV.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Civil Engineering), Semester-III

25CEU3CC8T: Surveying & Geomatics

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
		POE	25 Marks
Introduction:			
This course covers principles of limit state design for structural steel members, various types of rolled sections and its classifications. The curriculum includes the design of connections, tension members, compression members, beams, and column bases. Introduction to pre-engineered buildings.			
Course Prerequisite:			
This course requires knowledge of Engineering Mathematics and Basic Civil Engineering			
Course Objectives:			
1. Understand principles of accurate field data collection, recording, and reduction using conventional and electronic instruments. 2. Develop skills in operating modern surveying equipment safely and collaboratively in teams. 3. Apply advanced geospatial technologies (GNSS, Remote Sensing, GIS) for measurement, mapping, and analysis in Civil Engineering projects. 4. Prepare survey plans, maps, reports, and perform computations for areas, volumes, curves, and layouts.			
Course Outcomes:			
After completing the course, students will be able to 1. Perform temporary adjustments and operate Auto Level, Total Station, DGPS/GNSS, and modern instruments for accurate measurements of distances, levels, angles, and coordinates. 2. Conduct leveling, contouring, and route surveying projects, including data collection, reduction, and preparation of plans/profiles/cross-sections. 3. Apply modern geospatial techniques, including GPS/GNSS positioning modes, Remote Sensing (Including drone/UAV), and GIS for data processing and Civil Engineering applications. 4. Use open-source/academic software for map preparation, digitization, analysis, and reporting of surveying projects, including area/volume calculations and layout designs.			
Unit – I	Levelling Instruments and Applications	6 Hours	



Levels: Construction, temporary adjustments, errors, and use of Auto Level, Digital Level, and Tilting Level. Methods of levelling: Differential, profile, reciprocal, trigonometric. Contouring: Direct and indirect methods, interpolation techniques, characteristics and uses of contour maps in Civil projects.		
Unit – II	Modern Surveying Instruments	6 Hours
Electronic Distance Measurement (EDM), Laser Levels, and Digital Levels: Principles and applications. Total Station: Types, construction, temporary adjustments, working principles, data collection, and onboard calculations (distance, angles, elevation differences).		
Unit – III	Global Navigation Satellite Systems (GNSS/GPS)	6 Hours
GNSS Principles: Trilateration, segments (Space, Control, User), satellite features, observables. Receivers: Navigation, surveying, geodetic types. Positioning methods: Absolute, Relative/Differential GNSS, Real-Time Kinematic (RTK), Static, Rapid Static. Applications in Civil Engineering: Control surveys, precise positioning, deformation monitoring.		
Unit – IV	Remote Sensing and Emerging Techniques	6 Hours
Photogrammetry: Terrestrial and aerial principles, scale, relief displacement, flight planning. Unmanned Aerial Vehicles (UAVs/Drones): Introduction, components, applications in surveying and mapping. Electromagnetic Remote Sensing: Spectrum, energy interactions, atmospheric effects, spectral reflectance, image resolution, interpretation. Applications in Civil Engineering: Land use, disaster management, infrastructure monitoring.		
Unit – V	Geographical Information System (GIS) and Project Applications	6 Hours
GIS: Components, spatial/non-spatial data, vector vs. raster models, data preprocessing (rectification, registration, interpolation). GIS analysis: Overlay, buffering, network analysis; open-source tools (e.g., QGIS). Project surveying: Building layout, route alignment, public works surveys, curve setting (horizontal/vertical). Computations: Areas, volumes, earthwork from cross-sections/topographic data.		
INTERNAL CONTINUOUS ASSESSMENT ICA (25 Marks) based on journal/field book, drawing sheets, and project submissions. Students must record observations in a field book and submit soft/hard copies. Use open-source software(e.g., QGIS for GIS, LibreCAD/FreeCAD for drafting). Note: Emphasis on team work, safety protocols, software integration, and real-world Civil applications. Suggested Experiments/Projects (Any 6; mandatory projects 3 &4): 1. Study and temporary adjustments/use of Auto Level. 2. Study and use of Total Station (distance, angle, elevation measurements). 3. Project: Contour map preparation (minimum 1 hectare area) using block contouring/direct method (data via Auto Level/Total Station). 4. Project: Route surveying (minimum 1 km) for alignment/profile/cross-sections (data via Level/Total Station). Submit data in CSV/Excel (soft copy), drawings using open source/academic drafting software (individual submission). 5. Remote Sensing: Study of Mirror Stereoscope (air base), Parallax bar for height differences; satellite image interpretation. 6. GNSS/GPS: Field data collection using surveying/mapping receiver.		



7. GIS: Geo-registration, digitization, vector/raster analysis of a map; small GIS project with processing steps and report (hard copy prints).

Text Books

1. Madhu, N, Sathikumar, R, and Satheesh Gobi, Advanced Surveying: Total Station, GIS, and Remote Sensing, Pearson India, 2006.
2. Manoj, K. Arora, and Badjatia, Geomatics Engineering, Nem Chand & Bros, 2011
3. Bhavikatti, S.S., Surveying and Levelling, Vol. I and II, I.K. International, 2010
4. Chandra, A.M., Higher Surveying, Third Edition, New Age International (P) Limited, 2002.
5. Arora, K.R., Surveying, Vol-I, II, and III, Standard Book House, 2015.

Reference Books

1. Sharma, J.L. – Advanced Surveying, CBS Publishers.
2. Lillesand, T.M., Kiefer, R.W. – Remote Sensing & Image Interpretation, John Wiley & Sons.
3. Lo, C.P., Yeung, A.K.W. – Concepts and Techniques of GIS, Prentice Hall.
4. Chang, Kang-tsung – Introduction to GIS, Tata McGraw-Hill.
5. Reddy, M. Anji – Remote Sensing and Geographical Information System, B.S. Publications.

e-Resources:

1. NPTEL – Surveying (Basic & Advanced) – <https://nptel.ac.in/> (relevant modules on Total Station, GPS, Remote Sensing).
2. IIRS ISRO e-Learning – Geomatics & Remote Sensing Courses – <https://elearning.iirs.gov.in/>.
3. Open resources: QGIS tutorials, USGS EarthExplorer for satellite data, Drone surveying guidelines.



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Civil Engineering), Semester-III****25CEU3FP9L: Community Engagement Project/ Field Project**

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	-
Practical	4 Hours/week	ISE	-
Credits	2	ICA	50 Marks
Introduction:			
Community Engagement Project/ Field Project is an experiential learning strategy that integrates meaningful community engagement with instruction, participation, learning, and community development. It applies the experience to personal and academic development. It is meant to link the community with the institutes for mutual benefit. The community will benefit from the focused contribution of the students for the village/ local development. The institute finds an opportunity to develop social sensibility and responsibility among students and also to emerge as a socially responsible institution.			
Course Prerequisite:			
This course requires basic communication and interpersonal skills, understanding of social and environmental issues, teamwork and collaboration skills, report writing and documentation skills.			
Course Objectives:			
1. To sensitize the students to the living conditions of the people who are around them. 2. To bring about an attitudinal change in the students and help them to develop societal consciousness, sensibility, responsibility, and accountability 3. To make students aware of their inner strength and help them to find new /out of out-of-the-box solutions to the social problems 4. To make students socially responsible citizens who are sensitive to the needs of the disadvantaged sections 5. To help students to initiate developmental activities in the community in coordination with public and government authorities.			
Course Outcomes:			
After completing the course, students will be able to 1. Apply the knowledge to solve real-world problems 2. Demonstrate the complexity of understanding, problem analysis, problem-solving, critical thinking, and cognitive development 3. Develop interpersonal skills, particularly the ability to work well with others, and build leadership and communication skills.			



4. Improve social responsibility and citizenship skills.
5. Develop connections with professionals and community members for learning and career opportunities.

Procedure:

- Form a group of not more than 5 students.
- A mentor/guide will be allotted for each group.
- Students should finalize a particular habitation or village, or municipal ward, as far as possible, in the near vicinity of their place of stay.
- Students may work in close association with Non-Governmental Organizations like Lions Club, Rotary Club, etc., or with any NGO actively working in that habitation.
- Then, they should conduct a preliminary survey including the socio-economic conditions of the allotted habitation, in terms of their own domain or subject area. Or it can even be a general survey, incorporating all the different areas.
- If required, a survey form based on the type of habitation (rural, urban etc.) should be prepared before visiting the habitation.
- The Governmental agencies, like revenue administration, corporation and municipal authorities, and village secretariats may be aligned for the survey.
- Analysis of the collected data should be done.
- A solution should be proposed for the problem identified.

Students should prepare a report that includes the following points.

- Introduction
- Primary Data obtained through a survey/ field visit
- Analysis of collected data
- Proposed Solution

Students may take help from different government departments like –

- Agriculture
- Health
- Marketing and Cooperation
- Animal Husbandry
- Horticulture
- Fisheries
- Sericulture
- Revenue and Survey
- Natural Disaster Management
- Irrigation
- Law & Order
- Excise and Prohibition



- Mines and Geology
- Energy

Examples of community engagement/field projects are as below:

- Study of per capita domestic water consumption in the selected colonies in the ward
- Study and characterization of domestic waste generation in the ward
- Analysis of depot-level operations data
- Study of depot-level maintenance processes
- Study and mapping of open drains in the ward
- Study of the availability and access of public toilets in the ward
- Study and mapping of community spaces in the ward



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Civil Engineering), Semester-IV****25CEU4CC1T: Environmental Engineering**

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
		POE	25 Marks
Introduction:			
The course covers the study of water and wastewater management and design practices. Key topics include identification of water resources, water quality checking, design of water supply schemes, and introduction to various wastewater treatment technologies.			
Course Prerequisite:			
This course requires the Basic knowledge of Engineering Chemistry and Fluid Mechanics			
Course Objectives:			
1. To make students describe the importance of water quality and quantity, and interpret the design of water supply systems 2. To make students apply the appropriate technologies for water treatment. 3. To make students explain the impacts of sewage and select design conveyance systems for sewage and stormwater. 4. To make students identify and apply the appropriate process for sewage treatment and sludge management.			
Course Outcomes:			
After completing the course, students will be able to			
1. Examine and explain various sources of water, water quality, and quantity estimation. 2. Analyze appropriate water treatment technology. 3. Interpret the concepts of designing and managing water supply systems. 4. Describe the fundamentals of wastewater generation, conveyance systems, wastewater quality, and illustrate various methods of wastewater treatment.			
Unit – I	Water & Quantity Estimation		4 Hours
Sources of water, water quality requirements for different applications, water quality standards, Environmental policy and legislation, water quality indices, water demand, population forecasting methods.			



Unit – II	Water Treatment Processes (Aeration, Mixing, and Settling)	6 Hours
Treatment: Philosophy, Unit processes, and operations Aeration: Process, Types of aerators, Design of cascade aerator, Coagulation: Physics and chemistry, Practice, Design of rapid mixer Flocculation: Theory, Design of slow mixer, Settling: Theory, Types, and Design of rectangular and circular clarifiers for type 1 settling, Design of clariflocculator		
Unit – III	Water Treatment Processes (Filtration and Disinfection)	6 Hours
Granular Filtration: Classification, Components of deep bed filter, clean filter bed headloss, Filter operation, Design of filter, Disinfection: Types, Ideal and non-ideal disinfectant, Chlorination, Chemistry of chlorination, Chlorine demand, Chlorination practice, UV and Ozone disinfection		
Unit – IV	Advanced Treatment Processes	4 Hours
Membrane filtration: Types, Basic concepts, Applications, Adsorption: Introduction, Basics of Carbon adsorption, Ion Exchange: Theory, Design of softener, Point of use purifiers		
Unit – V	Water Distribution Systems	4 Hours
Components of the water supply system, conveyance of water, distribution system, and water supply appurtenances, service reservoirs, and design.		
Unit – VI	Sewage & Sewage Treatment	6 Hours
Need for conveyance and treatment of sewage, domestic wastewater, and storm water estimation, conveyance of sewage - sewers, shape design parameters, Physico-chemical and biological treatment aerobic and anaerobic treatment systems, suspended and attached growth systems		
(A) List of Experiments (Any 10): Determination of: 1. pH value 2. Acidity 3. Alkalinity 4. Hardness 5. Solids-Total, Suspended, dissolved, volatile, and fixed 6. Chlorides 7. Residual Chlorides 8. Sulphates 9. Turbidity 10. Dissolved Oxygen		



11. Most Probable Number 12. Oil & Grease (B) Six assignments per unit, where students are required to develop wastewater treatment plant (WWTP) applications for real-world scenarios, along with a report on the wastewater treatment units at the plant visited.
Text Books
1. Engineering (Environmental Engineering I), Standard Book House, 6 th Edition, 2018. 2. Nathanson, J. A., “Basic Environmental Technology”, PHI Learning Private Limited, 5 th Edition, 2009. 3. Modi, P. N., “Wastewater Engineering” Standard Book House, 6 th Edition, 2018. 4. Peavy H, S, Rowe D, R, and Tchobanoglous G, “Environmental Engineering”, McGraw-Hill Book Company, Indian Edition, 2017.
Reference Books
1. Raju, B.S.N., “Water Supply and Wastewater Engineering,” Tata McGraw-Hill Private Limited, New Delhi, 2nd Edition, 2000. 2. Garg, S.K.. “Water Supply Engineering” , Khanna Publishers, 33rd Edition, 2010. 3. Modi, P. N., “Water Supply Engineering (Environmental Engineering I)” , StandardBook House, 6th Edition, 2018. 4. Nathanson, J. A., “Basic Environmental Technology” , PHI Learning Private Limited, 5th Edition, 2009. 5. Modi, P. N., “Wastewater Engineering” Standard Book House, 6th Edition, 2018. 6. Peavy H, S, Rowe D, R, and Tchobanoglous G, “Environmental Engineering” ,McGraw-Hill Book Company, Indian Edition, 2017.
e-Resources
1. NPTEL, Course: Water Supply Engineering



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Civil Engineering), Semester-IV****25CEU4CC2T: Concrete Technology & Material Testing**

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction:

This course provides basic knowledge regarding the physical and chemical properties of various ingredients of concrete. It deals with testing procedures and standard specifications for checking the suitability of ingredients for making good-quality concrete. Also provides fundamental knowledge regarding the strength and durability properties of fresh and hardened concrete. It explains the procedure and gives hands-on experience in concrete mix design for various concrete grades using relevant Indian Standard (IS) codes. The course provides exposure to a variety of established material testing procedures and techniques, according to Indian Standards to check the suitability of materials in construction.

Course Prerequisite:

Primary knowledge of Basic Civil Engineering and building materials.

Course Objectives:

1. To make students aware of the physical and chemical properties of the ingredients of concrete.
2. To make students learn the behavior of fresh and hardened concrete
3. To make students design the concrete mixes for different grades using professional codes.
4. To make students apply quality control techniques in the field
5. To make students carry out Material Testing using relevant codes in the laboratory

Course Outcomes:

After completing the course, students will be able to

1. Relate material characteristics and their influence on the properties of concrete.
2. Distinguish concrete behaviour based on its fresh and hardened properties.
3. Illustrate the proportioning of different types of concrete mixes for required fresh and hardened properties using Indian standard codes.
4. Apply quality control techniques in the field
5. Carry out Material Testing in the laboratory



Unit – I	Ingredients of Concrete	8 Hours
Cement manufacturing process, chemical composition, and their importance, hydration of cement, types of cement. Testing of cement. Fine aggregate: Functions, requirements, Alternatives to River sand, Coarse aggregate: Importance of size, shape, and texture. Grading and blending of aggregate. Testing on the aggregate requirement. Water– qualities of water. Chemical admixtures – plasticizers, superplasticizers		
Unit – II	Fresh Concrete	6 Hours
Factors affecting workability. Measurement of workability–slump, Compaction factor, Vee- Bee Consistometer, flow tests. Segregation and bleeding. Process of manufacturing of Concrete- Batching, Mixing, Transporting, Placing, and Compaction. Curing – Methods of curing – Water curing, membrane curing, steam curing, accelerated curing. Good and bad practices of making and using fresh concrete.		
Unit – III	Hardened Concrete	6 Hours
Factors influencing strength, W/C ratio, gel/space ratio, and accelerated curing. Testing of hardened concrete, Creep –factors affecting creep. Shrinkage of concrete –plastic shrinking and drying shrinkage, Factors affecting shrinkage. Definition and significance of durability. Factors influencing durability, Durability requirements as per IS-456, latest version.		
Unit – IV	Concrete Mix Proportioning	7 Hours
Concept, variables in proportioning and exposure conditions, Selection criteria of ingredients used for mix design, and Methods of Concrete mix Proportioning, Examples of Mix Proportioning using IS-10262:2019. Quality control of concrete – Factors causing variations, field control.		
Unit – V	Testing of Materials	3 Hours
Tension test on Mild and Tor Steel, Stress-strain relationship, Water absorption, and Compression test on burnt Bricks.		
INTERNAL CONTINUOUS ASSESSMENT (ICA) A. Testing of cement Testing of cement viz. Consistency, Fineness, setting time, Soundness, and Compressive Strength, using relevant IS Codes B. Testing of aggregates - Specific Gravity & Water Absorption of Coarse Aggregate & Fine Aggregate - Sieve analysis of Coarse Aggregate & Fine Aggregate - Bulk density of Coarse Aggregate & Fine Aggregate - Bulking of Fine Aggregate - Silt Content of Fine Aggregate C. Tests on Concrete: - Slump cone test - Compaction Factor test - Vee- Bee Consistometer test - Flow Table test - Compressive Strength Test, Flexural Strength, and Split Tensile Strength		



D. Concrete Mix design: Proportioning as well as Experimental, as per IS10262: 2019, shall be demonstrated. E. Testing of materials: - Tension test on Mild and Tor Steel - Water absorption and Compression test on burnt Bricks
Text Books - Concrete Technology (Theory and Practice) by Shetty, M. S., & Jain, A. K. S. Chand Publishing. - Concrete Technology by Neville, A. M., & Brooks, J. J. England: Longman Scientific & Technical. - Concrete Technology by S.S.Bhavikatti. Wiley Publishing - Concrete Technology by A.R. Santhakumar. Oxford University Press, New Delhi. - Concrete Technology (Theory and Practice) by Gambhir, M. L. McGraw-Hill Editions.
Reference Books - Concrete Technology by Job Thomas, CENGAGE Learning. - Highway Materials & Pavement by Khanna & Justo, Nem Chand & Broths. - Handbook on Advanced Concrete Technology by Nayak, N. V., & Jain, A. K. Alpha Science International. - Concrete-Microstructure, Property and Materials by Kumar Mehta. P and Paulo J.M. Monteiro. McGraw-Hill Education. - I.S.456-2000 (reaffirmed 2021). Indian Standard Code of Practice for Plain & Reinforced Concrete. - I.S.10262-2019. Guidelines for Concrete Mix Design. - I.S.4031-1996 (reaffirmed 2021). Methods of Physical Tests on Cement - I.S.269-2015. Specifications for 33 Grade cement - I.S.8112-2013. Specifications for 43 Grade cement - I.S.12269-2013. Specifications for 53 Grade cement - I.S.383-2016 Specifications for Coarse & Fine Aggregates for Concrete. - I.S.1199-2018 Methods of Sampling & Analysis of Concrete. - I.S.2386-1963(reaffirmed 2021) Methods of Tests for Aggregates for Concrete - I.S.516-1959 (reaffirmed 2018) Methods of Tests for Strength of Concrete - Relevant I.S. codes for Testing and specifications of different building materials.
e-Resources - NPTEL – Concrete Technology- https://nptel.ac.in/courses/105/104/105104030 - IIT Madras – Material Testing and Evaluation-https://nptel.ac.in/courses/112/106/112106293 - YouTube – Concrete Technology Playlist by Civil Engineering Academy https://www.youtube.com/playlist?list=PL1Lr3QkAHS4rJ_0u-Qm97qtJZ9WeN1CSg





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Civil Engineering), Semester-IV

25CEU4CC3T: Structural Mechanics-II

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Tutorial	1 Hour/week	ISE	40 Marks
Credits	3	ICA	25 Marks
Introduction:			
This course covers structural analysis concepts, Eccentric loading on short columns, chimneys under wind pressure, Euler's and Rankine's formulas for columns, beam deflection methods, arch analysis, indeterminate structures, stiffness concepts, moment distribution, and the development of flexibility and stiffness matrices for beams and portal frames.			
Course Prerequisite:			
The student has completed a course in structural Mechanics-I. Understanding of statics, mechanics of materials, and engineering mathematics. Familiarity with stress, strain, equilibrium, internal forces, and basic structural design concepts is essential.			
Course Objectives:			
1. To analyse the behaviour of short columns under eccentric loading, consider the kern of the section and evaluate the effect of lateral pressures such as wind on slender structure like chimney on base. 2. To apply Euler's and Rankine's theories to analyse the behaviour of long columns, considering effective length, slenderness ratio, and limitations of the formula. 3. To apply Moment area, conjugate beam, and strain energy methods to determine slope & deflection in beams, analyse three-hinged arches for, radial shear, and normal thrust. 4. To Analyse and evaluate indeterminate structures by understanding degrees of indeterminacy, stiffness concepts, and applying the moment distribution methods for beams and portal frames. 5. To apply stiffness and flexibility concepts including matrix formulations, to analyse beams and portal frames with $(DSI \leq 3)$.			
Course Outcomes:			
After completing the course, students will be able to 1. Determine stresses in short columns under eccentric loading, identify safe load positions within the kern, and analyse the effect of wind pressure on chimneys base. 2. Calculate the crippling load on long columns using Euler's and Rankine's formulas and evaluate column behaviour based on effective length and slenderness ratio.			



3. Compute slopes and deflection of determinate beams and analyse three hinged arches to determine thrust and internal forces. 4. Determine member forces and moments in indeterminate beams and non-sway portal frames using moment distribution method and assess structural behaviour based on degrees of indeterminacy. 5. Formulate and solve flexibility and stiffness matrices to determine member forces and moments in beams and portal frames with $(DSI \leq 3)$.		
Unit – I	Combined Direct and Bending Stresses	5 Hours
Eccentric loading on short columns, Kern of a section, load Eccentricity about both axes of the section. Chimneys subjected to wind pressure		
Unit – II	Behaviour of Axially Loaded Long Columns	3 Hours
Behaviour of axially loaded long columns – Effective length, Slenderness ratio. Crippling load by Euler's and Rankine's formula, assumptions, limitations.		
Unit – III	Slope and Deflection of Determinate Beams	4 Hours
Moment area method, Conjugate beam method. Deflection of beams by the strain energy method		
Unit – IV	Three-Hinged Arches	3 Hours
Concepts, types of arches, analysis of parabolic arches with supports at the same and different levels, and semi-circular arches. Determination of horizontal thrust, radial shear, and normal thrust.		
Unit – V	Introduction of Indeterminate Structures	3 Hours
Concept of Indeterminate structures, Degree of Static and Kinematic indeterminacy, Degrees of freedom for various types of structures, Methods of analysis, and comparison of force and displacement methods.		
Unit – VI	Moment Distribution Method	4 Hours
Concept of stiffness of a member, Relative stiffness, Distribution factors, the concept of moment distribution, Application to beams, and non-sway portal frames.		
Unit – VII	Flexibility method for beams and frames	4 Hours
Derivation of flexibility equation, flexibility coefficients, Development of flexibility matrix, Analysis of beams and portal frames $(DSI \leq 3)$.		
Unit – VIII	Stiffness Method for Beams and Frames	4 Hours
Concept of stiffness, linearly elastic structures, derivation of Stiffness equation, Stiffness Coefficients, Development of stiffness matrix, Analysis of beams $(D.K.I. \leq 3)$, and Analysis of Portal frames $(D.K.I. \leq 3)$.		
INTERNAL CONTINUOUS ASSESSMENT ▪ Consists of at least one assignment on each unit: ▪ The results of a few assignment problems are to be verified by using application software		



Text Books
1. Mechanics of Structures (Vol. II) – by S.B. Junnarkar, Charator Book Publishing House. 2. Structural Analysis- by Negi and Jangid, Tata McGraw-Hill Publishing Company Ltd., New Delhi 3. Analysis of Structures (Vol. II) – by Vazirani and Ratwani, Khanna Pub., Delhi 4. Structural Analysis- Matrix Approach- by Pandit & Gupta, Tata McGraw-Hill Publishing Company Ltd., New Delhi 5. Structural Analysis – II by Bhavikatti, Vikas Publications, New Delhi 6. Structural Analysis, by R. Agor, Khanna Publishing House 7. Mechanics of Materials, by B.C. Punmia & A.K. Jain, Laxmi Publications 8. Advanced Structural Analysis, by A.K. Jain, Nem Chand Bros. 9. Theory of Structures, by Punmia, Laxmi Publications
Reference Books
1. Structural Analysis by C. S. Reddy, Tata McGraw-Hill. Graw Hill, New Delhi. 2. Matrix Analysis of Structures- by Gere and Weaver, CBS Publishers, New Delhi 3. Indeterminate Structural Analysis-by C. K. Wang, Tata McGraw-Hill Publishing Company Ltd., New Delhi 4. Theory of Structures- by Timoshenko & Young, Tata McGraw-Hill Publishing 5. Structural Analysis-Sixth Ed., by R. C. Hibbeler - Dorling Kindersley (India) Pvt. Ltd., Pearson Education, New Delhi
e-Resources
• https://www.youtube.com/playlist?list=PLOQ_D0oq27oCPBudQXnRf8G4UNQAOpEcV





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Civil Engineering), Semester-IV

25CEU4EM4T: Principles of Management and Quantitative Technique

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Credits	2	ISE	40 Marks

Introduction:

This course provides a thorough introduction to and application of the decision-making techniques required for decision-making in an organization. The course also introduces material and material management concepts and applies inventory control techniques for effective decision-making. Additionally, the course covers Lean Construction and ERP, along with the application of quality control charts for monitoring and checking.

Course Prerequisite:

Basic knowledge of statistics, and activities in civil engineering constructions

Course Objectives:

1. To make students aware of the function of management planning required for the successful functioning of an organization.
2. To teach students various decision-making techniques.
3. To instruct students on material management techniques and inventory control methods in the construction industry.
4. To educate students about Lean Construction and Construction ERP.
5. To teach students statistical methods and control charts

Course Outcomes:

After completing the course, students will be able to

1. Demonstrate the function of management for effective management of construction projects.
2. Apply various decision-making techniques in the construction industry.
3. Identify inventory costs associated with construction projects and apply ABC analysis and EOQ for inventory control, along with break-even analysis for decision-making in the construction industry.
4. Describe lean construction techniques and ERP in the construction industry.
5. Use statistical methods and control charts (X, R, p, c charts) for quality control of materials and workmanship in civil engineering projects.

Unit – I	Functions of Management:	4 Hours
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Functions of Management: Planning, Organizing, Staffing, Leading, Directing, Monitoring, and Controlling Management by Objectives (MBO). Organization and its types: Formal and informal organization, Centralization, Decentralization, Line organization, Line and staff organization, and Functional organization.		
Unit – II	Decision Making and Its Types	10 Hours
Decision under certainty: Linear Programming, Formulation of simple L-P model, Graphical method, Simplex method, Duality. Application of Linear Programming in ‘Transportation Problems: North-West corner method, least cost method, Vogel’s Approximation method (Only Initial Basic Feasible Solution), and Application of Linear Programming in ‘Assignment problems		
Unit – III	Decision under uncertainty and Risk:	4 Hours
Decision under Uncertainty: Wald’s (Principle) Criteria, Savage (Principle) Criteria, Hurvitz (Principle) Criteria, and Laplace (Principle) Criteria Decision under Risk: Expected Monetary Value (EMV) and Expected Opportunity Loss Value		
Unit – IV	Inventory control	6 Hours
Introduction, inventory cost, EOQ analysis, ABC analysis, and safety stocks. Break-even analysis		
Unit – V	Lean Construction and Construction ERP:	4 Hours
Introduction to Lean Construction and Construction ERP		
Unit – VI	Statistical Quality Control:	4 Hours
Introduction: Statistical Quality Control, Control chart and types: X, R, P, C charts		
INTERNAL CONTINUOUS ASSESSMENT (ICA) The Internal Continuous Assessment (ICA) shall consist of a minimum of six assignments based on the entire curriculum, each with a weightage of 10 marks.		
Text Books		
1. A Textbook of Organizational Behaviour, CB Gupta, S. Chand Publications 2. Construction Engineering & Management, S.C. Sharma & S.V. Deodhar, Khanna Book Publishing 3. Optimization Techniques, S.S. Rao, Wiley Eastern India 4. Operation Research, Hamdy A. Taha, Operation Research, Prentice Hall of India, New Delhi, 8thEd.2011 5. Store Management, Menon K. S., Store Management, McMillan Co. New Delhi, 2nd Ed.		



1998.

6. Statistical Quality Control, E. L. Grant, Statistical Quality Control, Wiley International Education, 6th Ed.
7. Udo Linden, Mrunalini Kulkarni, Hit-Office Construction ERP technical manual, Engineering Design Software and Services Pvt. Ltd., Pune, April 2018 Edition.

Reference Books

1. Total Quality Management, Ponia & Sharma, Khanna Publishing House, Delhi
2. Engineering Management: Industrial Engineering & Management, S.C. Sharma, Khanna Publishing House, Delhi
3. Principles and Practice of Management, Prasad, L.M, Sultan Chand
4. Organizational Behaviour, L.M. Prasad, Sutan Chand and Sons.
5. Handbook of Construction Management, Joy PK, Macmillan
6. Construction Project Management, Jha, Pearson
7. Total Quality Management, Gopal, PHI Publications
8. Industrial Engineering & Operations Management, S.K. Sharma. S.K. Kataria & Sons
9. Principles of Operation Research: Prentice Hall of India, 2nd Ed. 1925, Wagner H. M.
10. Operations Research: Shaum's outline series, Richard Bronson Govindsami N., Tata McGraw-Hill, 2nd Ed., 2004
11. Material Management, Gopal Krishnan, Sudeshan,
12. Handbook of Quality Control, Juran J. M., A. B. Godfrey, McGraw-Hill

e-Resources

1. NPTEL – Principles of Management (IIT Kharagpur)- <https://nptel.ac.in/courses/110/105/110105146/>
2. NPTEL – Quantitative Finance and Operations Research (IIT Madras)- <https://nptel.ac.in/courses/110/106/110106062/>
3. Linear Programming Solver (Open Solver for Excel)- <https://opensolver.org/>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (All Branches), Semester-IV

25CMU4AE7T: General Proficiency

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ICA	50 Marks
Tutorial	1 Hour/week		
Credits	2		

Introduction:

In today's global competitive professional setting, engineers need to possess a wide range of skills to succeed beyond technical knowledge. They must demonstrate strong communication, problem-solving, interpersonal and leadership skills. The General Proficiency course is strategically designed to facilitate the transition from academic settings to industry, emphasizing the holistic development of students. Its structured, experiential approach enhances essential employability skills and promotes personal growth, responsible citizenship, and well-being. Key focus areas include professional communication, soft skills, leadership development, and social skills. By completing this course, students are better prepared to navigate campus placements and resolve the challenges of the professional world, fostering their ethical responsibility and social awareness.

Course Objectives:

1. To understand job-specific ATS-friendly resume writing and effective email communication.
2. To develop structured presentations and effective delivery techniques using AI tools.
3. To understand the importance of group discussions and increase effective participation strategies.
4. To strengthen self-presentation and interview skills using verbal and non-verbal strategies.
5. To write clear and well-structured notices, agendas, and minutes of the meeting

Course Outcomes:

After completing the course, students will be able to

1. Create job-specific ATS-friendly resumes and write clear, professional emails.
2. Design and deliver confident, well-organised presentations using AI support.
3. Participate actively in group discussions using effective communication skills.
4. Face an interview with confidence by applying personal interview techniques.
5. Write well-structured notices, agendas, and correct minutes of meetings in formal language.



Unit – I	Professional Communication:	4 Hours
ATS-friendly resume preparation, Job specific resume, Tips to write an effective resume, Do's and don'ts of professional email writing etiquette		
Unit – II	Presentation Skills	4 Hours
Introduction to presentation, Types of Presentation, Importance of presentation Skills for Engineers, Structure of effective presentation, Use of AI for making effective PPT, 4 P's of Effective presentation		
Unit – III	Group Discussion (GD) Skills	4 Hours
Role of group discussion in placement selection process, Steps of GD, Tips for effective GD preparation, Dos and don'ts of effective group discussion. Body language and time management		
Unit – IV	Personal Interview Techniques	2 Hours
Introduction to interview, How to prepare for interview, Personal appearance, Body language, communication skills, professionalism, SWOC Analysis, STAR method, interview questions (FAQs), Effective strategies for answering, Closing interview		
Unit – V	Notices, Agendas, and Minutes of Meeting	2 Hours
Writing notices, Agenda, format and outline, formal language, Sample notices for meetings, events, seminars, etc. Structuring effective 'minutes of meeting', Do's and don'ts of minutes of meeting.		
Unit – VI	Personality Development for Engineering Students	2 Hours
Types of personalities, Personality traits, Personal grooming and Body language for Success, Personality depiction, Time management and discipline, Professional etiquettes.		



INTERNAL CONTINUOUS ASSESSMENT (ICA)

List of assignments: Students should be completed at least 5 writing and 5 oral assignments.

- 1) **Resume Creation:** Students will prepare a one-page ATS-friendly resume tailored to their job description, ATS Score Checker
- 2) **Email and Letter Writing:** Student will follow email etiquette and write cover letter for placement opportunity and email an attachment of resume and cover letter
- 3) **Writing Notice, Agenda, and Minutes of Meeting:** Write the Notice, Agenda, and Minutes of meeting in a business and professional setting
- 4) **Prepare PPT file** on the topic related to engineering / current trends / social issues to present it.
- 5) **Public Speaking:** Extempore speech, Describe a situation.
- 6) **Group Discussion:** Students will participate in GDs on current topics, and after the GD, the performance of participants will be reviewed by a peer assessment method.
- 7) **Personal Interviews:** Students will undergo mock HR, simulations in front of peers and instructors, followed by personalized feedback on performance and body language.
- 8) **SWOT Analysis of Personality:** Each student will perform a personal SWOT analysis to identify their strengths, weaknesses, opportunities, and threats in the context of personality development and employability.
- 9) **Leadership and Teamwork Exercise:** Solving and discussing Self and Business Case studies

Suggested Components for Oral Activity:

1. Oral Presentation (Individual)

- Topic related to engineering / current trends / social issues.
- Duration: 3–5 minutes
- Focus: clarity, confidence, structure, language

2. Group Discussion (GD)

- Group of 6–8 students, Topics: technical, ethical, or general awareness
- Evaluation on participation, logic, teamwork

3. Extempore Speech/Think Pair and share

- Announcement of topic, Thinking time: 5 minutes
- Speaking time: 5 minutes

4. Mock Interview/Panel Interview

- Professional scenarios: interview, Interviewer- Interviewee interaction
- Personal interview, and panel interview,
- Technical + HR questions, SWOT Analysis, STAR, FAQs
- Resume-based interaction

5. Seminar / PPT Presentation

- Use of AI tools allowed (with ethics discussion)
- Slides-5–7, Time-5 minutes presentation

6. **Leadership and Team Skills:** Role and responsibilities of team leader and teamwork of Idea/ Problem Statement Project Competition, Problem solving and discussing Self and Business Case studies



Text Books

1. Soft Skills: An Integrated Approach to Maximize Personality, w/cd – Gajendra Singh Chauhan & Sangeeta Sharma, Paperback Publications: Wiley Publisher India Pvt. Ltd.2015.
2. Communication Skills for Professionals – Nira Konar, PHI Learning, 3rd Edition, 2022.
3. On Writing Well – William Zinsser, Harper Resource Book, 2001.
4. Technical English – Dr. M. Hemamalini, Wiley India Pvt. Ltd.
5. Professional Speaking Skills – Aruna Koneru, Paperback, January 2018.
6. Group Discussion and Interview Skills – Priyadarshi Patnaik, Cambridge University Press India, 2nd Edition, 2015

Reference Books

1. Soft Skills: Know Yourself and Know the World- Dr. K. Alex, Binding Paperback, S. Chand Publishing, 2017.
2. Soft Skills: A Textbook for Undergraduates – Ajay R. Tengse, Generic Publication, 2015.
3. Communication Skills – Sanjay Kumar & Pushpa Lata. Oxford University Press: Paperback 2024.
4. Managing Soft Skills for Personality Development – B. N. Ghosh, McGraw-Hill.
5. Personality Development and Communication Skill-I. Paperback, 2014.
6. Personality & Soft Skills Development. Rajeev Kumar. Perfect Paperback: 2023.
7. Soft Skills for Everyone with CD – Jeff Butterfield, Cengage Learning: Paperback, 2011.
8. Soft Skills for Managers – Dr. T. Kalyana Chakravarthi & Dr. T. Latha Chakravarthi, Biztantra Publication. Kindle Edition, 2011.

e-Resources

NPTEL – Soft Skills Development (IIT Kharagpur / IIT Madras)-

<https://nptel.ac.in/courses/109/105/109105115/>

SWAYAM – Life Skills for Engineers-https://swayam.gov.in/nd1_noc20_hs36/preview

https://youtu.be/rnqq_14DF0I?si=tlEz_P9VynBDatbh

https://youtu.be/rnqq_14DF0I?si=tlEz_P9VynBDatbh

<https://youtu.be/33P93jJuw0w?si=wYoZN3dI3ZQTTbno>

https://youtu.be/w_90Wi_XUaU?si=8ub4T126yzj7_AlC

<https://youtu.be/kGSmmSLhxjs?si=CAfJ6P9JjZ9Aciit>

https://youtu.be/nduK_MumEmY?si=1GmUhK9B2HA0N0qA

<https://youtu.be/haEjVWFnVgg?si=Wgy7RXEE3iDxIolg>

<https://youtu.be/jn9JL2L1Ung?si=wL9qFUU0EvJYSdfc>



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Civil Engineering), Semester-IV****25CMU4VE8T: Professional Ethics**

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ISE	50 Marks
Credits	2		
Introduction:			
This course is designed to explore the principles and standards of moral and ethical conduct in professional settings. This course aims to equip students with the necessary tools to navigate complex ethical dilemmas and make informed decisions that uphold the integrity and ethical standards of their profession. It emphasizes the importance of ethical behaviour in building trust, maintaining credibility, and fostering a positive professional environment.			
Course Prerequisite:			
This course requires basic communication and comprehension skills, general awareness of engineering practice, basic understanding of laws and regulations, reflective mindset and willingness to discuss real-world ethical scenarios.			
Course Objectives:			
1. To make students aware of Professional Ethics in engineering 2. To make students aware of various theories in Professional Ethics 3. To make students learn about the safety, risk, and responsibilities of an engineer 4. To make students learn about the global issues in Professional Ethics			
Course Outcomes:			
After completing the course, students will be able to 1. Follow Professional Ethics in his life. 2. Describe various theories in Professional Ethics. 3. Identify the safety, risk, and responsibilities of an engineer. 4. Behave consciously regarding global issues in Professional Ethics.			
Unit – I	Introduction to Professional Ethics	3 Hours	
Introduction, Engineering and Professionalism, Two models of Professionalism, Three types of morality, Preventive Ethics, Aspirational Ethics			
Unit – II	Engineering Ethics	4 Hours	
Senses of engineering ethics, Variety of Moral Issues, Types of Inquiry, Recent developments towards ethics in engineering, Moral Dilemmas-steps to solve moral dilemmas.			



Unit – III	Theories in Engineering Ethics	4 Hours
Kohlberg's Theory, Gilligan's Theory, Consensus and Controversy, Models of Professional Roles, Theories about Right Action, Self-interest, Customs and Religion, Uses of Ethical Theories.		
Unit – IV	Engineering as Social Experimentation	4 Hours
Engineering projects vs Standard projects, Engineers as responsible experimenters, code of ethics, Industrial standards.		
Unit – V	Safety and Risk	4 Hours
Concept of safety, Engineers and safety, Risk- Types of accidents, Risk Benefit analysis, Reducing risk, Risk Management.		
Unit – VI	Responsibilities of an Engineer	4 Hours
Collegiality, Loyalty, Respect of Authority, Collective Bargaining, Confidentiality, Conflict of Interest.		
Unit – VII	Rights of an Engineer	3 Hours
Professional Rights, Employee Rights, Whistle Blowing, Intellectual Property Rights, Discrimination, and Preferential Treatment.		
Unit – VIII	Global Issues	4 Hours
Multinational Corporation, Ways of promoting morally just measures, Environmental Ethics, Computer Ethics, Weapons Development, Engineers as Managers, Expert Witnesses and Advisors, Moral Leadership, Corporate Social Responsibility.		
Text Books		
1. R.S. Naagarazan, A Text Book of Professional Ethics & Human Values, New Age International, 2006. 2. Professional Ethics: R. Subramanian, Oxford University Press, 2015. 3. Dr. N. Venkateswaran, Professional Ethics in Engineering, Sree Kamalamani Publications.		
Reference Books		
1. Charles E. Harris Jr., Michael S. Pritchard, and Michael J. Rabins, Engineering Ethics: Concepts and Cases, 4th Edition. Delhi)		
e-Resources		
1. NPTEL – Professional Ethics (IIT Kanpur / IIT Madras) https://nptel.ac.in/courses/109104068 2. SWAYAM – Ethics in Engineering Practice https://swayam.gov.in → Search “Ethics” or “Professional Ethics”		





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Civil Engineering), Semester-IV

25CEU4CC9L: Building Planning and Modelling

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ICA	25 Marks
Practical	2 Hours/week	POE	25 Marks
Credits	2		

Introduction:

This course provides a thorough introduction to the functional requirements and structures of buildings. It also covers the development of CADD skills for designing and building components, creating detailed perspective view drawings, and planning and designing “Residential Buildings” or “Public Buildings” with CADD to ensure compliance with building bylaws

Course Prerequisite:

Basic functional requirement of Residential Buildings and Public Buildings, and Building Services.

Course Objectives:

1. To make students plan and design “Residential Buildings” or “Public Buildings”, develop their ability to design such buildings according to specified requirements, and ensure adherence to National Building Code norms and standards.
2. To make students prepare detailed permission drawings for “Residential Buildings” or “Public Buildings” to obtain building permission from the competent authority using a suitable Computer Aided Drawing and Drafting (CADD) application software.
3. To make students design building services layouts plan such as furniture requirements, electrification points, water supply, and drainage systems, by standard norms using CADD application software
4. To make students produce perspective and line drawings, create perspective drawings of the building and line plans of any two public buildings using CADD application software, demonstrating a comprehensive understanding of architectural representation.
5. To make students compile a comprehensive report, prepare a detailed report on the selected “Residential Buildings” or “Public Buildings”, encompassing design rationale, adherence to standards, and an overall project assessment.

Course Outcomes:

After completing the course, students will be able to

1. Plan and design a “Residential Buildings” or “Public Buildings” according to requirements, adhering to National Building Code norms and standards.
2. Prepare “Permission Drawing” for residential buildings or public buildings to obtain building permission from the competent authority by using a suitable CAAD application software.



3. Plan and design an appropriate building services layout for “Furniture requirement, Electrification points, Water supply, and Drainage System” for a building as per the standards norms by using a suitable ‘Computer Aided Drawing and Design’ application software. 4. Prepare “Perspective drawing of the building” and “Line plan of any two “Residential Buildings” and “Public Buildings” by using a suitable CAAD application software. 5. Prepare a report on the selected type of building		
Unit – I	Planning and Design of Residential Buildings	6 Hours
Bungalows (Detached), Twin bungalows (Semi-Detached), Rowhouses, Ownership flats, Apartments, Parking Area Criteria		
Unit – II	Planning and Design of Public Buildings	6 Hours
Educational Building: Pre-primary and primary school, Secondary and Higher Secondary school, Degree School (College). Institutional Building: - Health centers and Hospitals. Business and Mercantile building – Shops, banks, markets, & departmental stores. Office and Other buildings: Post office, administrative building, etc. Parking Area Criteria (for all above Public Buildings)		
Unit – III	Building Permissions and Procedure	3 Hours
Procedure and list of documents for Building Permission and significance of various certificates (Commencement Certificate, Plinth Completion Certificate, and Occupancy Certificate).		
INTERNAL CONTINUOUS ASSESSMENT (ICA) A. Preparation of drawings for any one “Residential Buildings” or “Public Building” by using AutoCAD 1. Building Permission Drawing 2. Building Permission Drawing Manual on Full Imperial sheet 3. Furniture layout plan 4. Electrification layout plan 5. Water supply and Drainage layout plan 6. Perspective drawing of the building. B. Report on building project under (A) above. C. Site visit for the type of building selected for planning and designing for the Internal Continuous Assessment (ICA) submission		



END SEMESTER EXAMINATION (ESE)

Practical Oral Examination:

1. Practical examination shall be based on an assessment of the knowledge of students about planning skills and CADD drafting skills related to residential buildings or public buildings. **(Maximum two hours shall be allotted to students to complete the given task on CADD during the Practical and Viva Exam.)**
2. In addition, an oral examination shall be based on the Practical and ICA.

Text Books

1. Building Construction: Arora and Bindra, Dhanpat Rai Publications
2. Building Design and Drawing – Y. S. Sane, Allies Book Stall
3. Principles of Perspective drawing- Shah, Kale, Patki, Tata McGraw Hill Publication Ltd, Delhi
4. Building Construction by Sushil Kumar, Standard Publishers Distributors, Delhi
5. Interior Design- Principles and Practice- M. Pratap Rao, Standard Publishers and Dist., Delhi
6. Building Planning and Design by Kumar Swami and Kameshwar Rao, Charotar Publishing House.
7. Civil Engg. Drawing- by M. Chakraborty, Published by M. Chakraborty – Kolkata
8. Civil Engineering Drawing – by R.S.Malik, G.S.Meo, Computech Publication Ltd, New Asian.
9. Learning 'AutoCAD' software –Omeru

Reference Books

1. Building Construction by McKay, W. B. & McKay, J. M. , Vol.III and IV, Donhead Publishing Limited
2. Modern Building Construction by Warland D. E., Vol. I and II, Pitman Publishing
3. Building Drawing – Shah, Kale, Patki, Tata McGraw-Hill Education
4. Built Environment by Shah, Kale, Patki, Tata McGraw-Hill Education
5. Construction science – by Edwin Walker, Selwyn Morgan, Hutchinson Educational
6. Time savers standards for buildings – Calendar Pub. McGraw Hill
7. Alternative Building Materials & Technology by Jagdish ,Reddy, Rao Published by New Age International, New Delhi.

e-Resources

NPTEL – Building Planning and Drawing (IIT Roorkee)
<https://nptel.ac.in/courses/105107123>



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Civil Technology), Semester-IV****25CEU4PC10P: Programming with Python**

Teaching Scheme		Examination Scheme	
Lectures	1 Hour/week	ICA	50 Marks
Practical	2 Hours/week	POE	25 Marks
Credits	2		
Introduction:			
Python is a popular, general-purpose, multi-paradigm, open-source scripting language. It is designed to emphasize code readability and has a clean syntax with high-level data types. It is suited for interactive work and quick prototyping while being powerful enough to write large applications. This course introduces the Python language, which has simple syntax, a powerful set of libraries, and a robust debugger and profiler.			
Course Prerequisite:			
Basic knowledge of programming and a programming language			
Course Objectives:			
1. Acquire knowledge of Python and the scripting language. 2. Acquaint with different tools in Python. 3. Understand and implement file-handling concepts in Python. 4. Write Python Programs for solving Civil Engineering problems.			
Course Outcomes:			
After completing the course, students will be able to			
1. Elaborate basics of Python programming. 2. Develop and implement control statements and functions with packages. 3. Apply NumPy and plotting tools in Python. 4. Write Python Programs for solving Civil Engineering problems.			
Unit – I	Introduction to Python:		4 Hours
Script Model Programming, Understanding Python variables, basic Operators, Declaring and using Numeric data types: int, float, complex, using string data type and string operations, defining list and list slicing, List manipulation using in-build methods, Use of Tuple data type, Dictionary manipulation.			



Unit – II	Python Program Flow Control, functions, and packages:	4 Hours
Conditional blocks using if, else, and elif, Simple for loops in Python, for loops using ranges, string, list, and dictionaries, Use of while loops in Python, Loop manipulation using pass, continue, break, and else. Programming using Python's conditional and loop blocks. Programming using string, list and dictionary in build functions. Organizing Python codes using functions, Understanding Packages, Powerful Lambda function in Python Programming using functions, modules, and external packages.		
Unit – III	NumPy and Matplotlib:	3 Hours
What is NumPy? How to install NumPy, Arrays, Array indexing, Array Vs Listing Data types, Array math, Broadcasting. Matplotlib -Plotting, subplots, and images		
Unit – IV	Python file operation:	4 Hours
Reading config files in Python, Writing log files in Python, Understanding read functions, read, readline, and readlines. Understanding write functions, write, and writelines. Manipulating the file pointer using seek. Programming using file operations.		
INTERNAL CONTINUOUS ASSESSMENT (ICA):		
Minimum 10 program assignments covering all the above topics.		
Text Books		
1. Gowrishankar, S., & Veena, A. (2018). Introduction to Python Programming. CRC Press (Taylor & Francis). 2. Kanetkar, Y. (2019). Let Us Python. BPB Publications. 3. Programming in Python 3, Mark Summerfield, Second Edition 4. Learning Python, Mark Lutz, O'reilly		
Reference Books		
1. Python Cookbook, David Beazley and Brian K. Jones, Third Edition, Shroff Publishers & Distributors Pvt. Ltd., ISBN :978-93-5110-140-6 2. Learning Python, Mark Lutz, 5th edition 3. Programming Python (English), Mark Lutz, 4th Edition 4. Testing Python, David Sale, Wiley India (P) Ltd., ISBN :978-81-265-5277-1		
e-Resources		
Python documentation - https://docs.python.org/3/		





ALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Second Year B.Tech. (Civil Technology), Semester-IV

25CMU4MC2T: Environmental Studies

Teaching Scheme

Lectures

1 Hour/week

Examination Scheme

ESE

50 Marks

Introduction:

The need for sustainable development is key to the future of mankind. Continuing the problems of all types of pollution, loss of forest, solid waste disposal, degradation of the environment, issues like economic productivity and national security, global warming, ozone layer depletion, and loss of biodiversity have made everyone aware of environmental issues. No citizen of the earth can afford to be ignorant of environmental issues. Environmental management has captured the attention of healthcare managers. Managing environmental hazards has become very important. It is now more critical than ever before for mankind as a whole to have a clear understanding of environmental concerns and to follow sustainable development practices. Destruction of habitats, overuse of energy resources, and environmental pollution are responsible for the loss of a large number of life forms. It is feared that a large proportion of life is which may be wiped out shortly.

Course Prerequisite:

This course requires basic understanding of science subjects from their higher secondary education, particularly in areas such as biology, chemistry, mathematics, physics, and geography. Familiarity with fundamental ecological concepts and general awareness about current environmental issues will be beneficial. Basic communication and analytical skills are also expected.

Course Objectives:

1. To introduce students to the multidisciplinary nature of environmental studies and the importance of environmental conservation for sustainable development.
2. To enable students to understand natural resources, ecosystems, biodiversity, and environmental pollution along with their causes, effects, and control measures.
3. To develop awareness about major environmental issues, related laws, and the role of individuals and society in addressing environmental challenges.

Course Outcomes:

After completing the course, students will be able to

1. Describe the significance of environmental studies, identify various natural resources and their associated problems, and suggest strategies for their sustainable use.



2. Analyze the structure and functions of ecosystems, assess biodiversity and its conservation methods, and discuss various types of pollution and their preventive measures. 3. Demonstrate understanding of key environmental policies, social issues related to the environment, and individual responsibility towards environmental protection and sustainable living.		
Unit – I	Nature of Environmental Studies:	2 Hours
Definition, scope, and importance. Multidisciplinary nature of environmental studies Needs for public awareness.		
Unit – II	Natural resources and associated problems:	2 Hours
a. Forest, resources, use, and over-exploration, deforestation, timber extraction, mining, dams, and their effects on forests and tribal people. b. Water resources, use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams' benefits and problems c. Mineral resources. And usage and exploitation, the environmental effects of extracting and using mineral resources. d. Food resources, world food problem, changes caused by agriculture effects of modern agriculture, fertilizer-pesticide problems e. Energy resources, growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources f. Land resources, land as a resource, land degradation man man-induced landslides, soil erosion, and decertification g. Role of individuals in the conservation of natural resources h. Equitable use of resources for a sustainable lifestyle		
Unit – III	Ecosystems:	2 Hours
Concept of an ecosystem a. Structure and function of an ecosystem b. Producers, consumers, and decomposers c. Energy flow in the ecosystem d. Ecological succession e. Food chains, food webs, and ecological pyramids Introduction types, Characteristics, features, structure, and function of the ecosystem: - a. Forest ecosystem b. Grassland ecosystem c. Desert ecosystem Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)		
Unit – IV	Biodiversity and its conservation:	2 Hours
a. Introduction-Definition, genetic, species, and ecosystem diversity b. Biogeographically classification of India c. Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic, and option values d. Biodiversity at global, national, and local levels		



e. India is a mega-diverse nation f. Western Ghats as a biodiversity region g. Hot spot of biodiversity h. Threats to biodiversity, habitat loss, poaching of wildlife, and man-wildlife conflicts i. Endangered and endemic species of India. j. Conservation of biodiversity, in-situ and Ex-situ conservation of biodiversity		
Unit – V	Environmental Pollution:	2 Hours
Definitions:- Causes, effects and control measures of a) Air Pollution b) Water pollution c) Soil Pollution d) Marine pollution e) Noise pollution f) Thermal pollution g) Near hazards h) Solid waste Management, causes effects and control measures of urban and Industrial wastes i) Role of an individual in presentation of pollution j) Pollution case studies k) Disaster management:-Floods, earthquake, cyclone and landslides, Tsunami		
Unit – VI	Social Issues and the Environment	2 Hours
a. From Unsustainable to Sustainable development b. Urban problems related to energy c. Water conservation, rainwater harvesting, watershed management d. Resettlement and rehabilitation of people, their problems and concerns e. Environmental ethics, Issues, and possible solutions f. climate change, Global warming, acid rain, Ozone layer depletion, nuclear accidents, and the holocaust. g. Consumerism and waste products		
Unit – VII	Environmental Protection:	3 Hours
Environment Protection Act a. Air (Prevention and Control of Pollution Act) b. Water (prevention and control of Pollution act) c. Wildlife Protection Act d. Population growth and human health, human rights		
Text Books		
1. Bharucha, E. – <i>Environmental Studies</i> (UGC-recommended, Universities Press). 2. Kaushik, A. & Kaushik, C.P. – <i>Environmental Science and Engineering</i> (New Age International). 3. Meenakshi, P. – <i>Environmental Science and Engineering</i> (PHI Learning). 4. Rao, P. Venugopala – <i>Principles of Environmental Science and Engineering</i> (PHI).		



5. **Gilbert M. Masters** – *Introduction to Environmental Engineering and Science* (Pearson Education).

Reference Books

1. Erach Bharucha (2013): Textbook of Environmental Studies for undergraduate courses, second Edition (2013).
2. P.S. Verna and V.K. Agarwal, 1983. Environmental biology, S. Chand Publications, New Delhi.
3. [https://www.google.co.in/ images](https://www.google.co.in/images)
4. <https://envfor.nic.in/legis/legis.html>
5. Dr Prakash Sawant (2009) “Environment studies” Fadake Publisher Kolhapur
6. Dr S. D Kadam (2005) “Human, Environment, and Pollution” Fadake Publisher Kolhapur
7. Environment studies- University Press, Solapur University, Solapur
8. Erach Bharucha-“Environmental Studies” UGE Press New Delhi
9. Dr. J.S. Samant (2005)-“Environmental Studies”, Shivaji University Press
10. Bharucha, E. (2004): Textbook for environmental studies for undergraduate students of all branches of higher education. University Grants Commission (UGC), New Delhi pp 249-286.

e-Resources

NPTTEL – Environmental Studies: For Undergraduate Students (IIT Kharagpur)

<https://nptel.ac.in/courses/120108004>

SWAYAM – Environmental Studies

<https://swayam.gov.in> → Search for “Environmental Studies”

